

ABSTRACT

The Music Streaming Data Analysis System is a Python-based interactive data analytics application developed to analyze, interpret, and visualize patterns in music streaming data. With the rapid growth of digital music platforms, large volumes of user and streaming-related data are generated, containing valuable information about listener preferences, popular artists, music genres, listening behavior, platform usage, and demographic characteristics. However, raw datasets are often difficult to interpret and require systematic processing to extract meaningful insights. The proposed system addresses this challenge by providing an interactive platform that transforms raw music streaming data into understandable analytical information. The system accepts music streaming datasets in Excel or CSV format and performs essential preprocessing operations such as data loading, validation, handling missing values, identifying duplicate records, correcting data types, and preparing the data for analysis. Pandas and NumPy are used for data manipulation, statistical calculations, grouping, filtering, and aggregation, while Plotly is used to create interactive visualizations such as bar charts for analyzing top artists, pie or donut charts for genre distribution, line charts for appropriate listening trends, and additional charts for platform, demographic, and listening behavior analysis. A Streamlit-based web interface provides an easy-to-use dashboard through which users can explore the dataset, apply filters, compare different categories, examine statistical summaries, and obtain dynamically generated insights based on the selected data. The system focuses on ensuring that all analytical results and visualizations are derived from the actual dataset rather than predefined or fabricated values. By integrating data preprocessing, exploratory data analysis, statistical analysis, interactive visualization, and insight generation into a single application, the proposed system demonstrates an end-to-end approach to practical data analytics using Python. The system can help users understand music consumption patterns more effectively and provides a foundation for future enhancements such as personalized music recommendations, predictive analysis, advanced listener segmentation, and time-series forecasting.